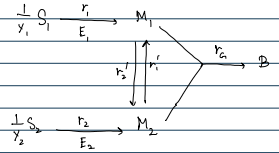
HW12

Tuesday, 18 April 2023 9:38 AM

Bioreactors, cybernetic modelling ; Anagrelastic

Question 1

(i) Product = Biomass ; M_1, M_2 → growth precursors ; Glucose, pyruvate → Substrate



Two elementary modes in the network:

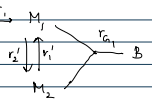
① Diauxic growth : $S_1 \rightarrow M_1$; $M_1 \rightarrow M_2$; $M_1 + M_2 \rightarrow B$ (Sequential growth)

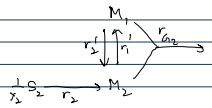
Preferential utilization of S_1 over S_2 ;

② Simultaneous utilization: $S_1 \rightarrow M_1$; $M_1 + M_2 \rightarrow B$

$S_2 \rightarrow M_2$

Two elementary modes are the individual diauxic reactions:





$\frac{1}{Y_1}S_1 \xrightarrow{r_1} M_1$

$M_1 \underset{r_2'}{\overset{r_1'}{\rightleftharpoons}} M_2$

$M_1 + M_2 \xrightarrow{r_{G_1}} B$

Elementary Mode 1

$\frac{1}{Y_2}S_2 \xrightarrow{r_2} M_2$

$M_1 \underset{r_2'}{\overset{r_1'}{\rightleftharpoons}} M_2$

$M_1 + M_2 \xrightarrow{r_{G_2}} B$

Elementary Mode 2

① Mode 1 (Diauxic growth) :

$$\frac{dc}{dt} = r_{G_1} v_1 + r_{G_2} v_2$$

$$\frac{ds_1}{dt} = -\frac{1}{Y_1} r_{G_1} Y_1$$

$$\frac{de_1}{dt} = \alpha_1^* + \frac{k_E S}{K_S + S} u_1 - \beta_e - e_1 (r_{G_1} + r_{G_2})$$

$$r_{G_1} = \frac{M_1 S_1 c}{K_{S_1} + S_1} \left(\frac{e_1}{e_{1,max}} \right)$$

$$r_{G_2} = \frac{M_2 S_2 c}{K_{S_2} + S_2} \left(\frac{e_2}{e_{2,max}} \right)$$

② Mode 2 (Simultaneous growth) ; $u_{S_1,M_1} = \frac{Y_{S_1,M_1} r_{S_1,M_1}}{Y_{S_1,M_1} r_{S_1,M_1} + r_{M_2,M_1}}$

u

(ii) Cybernetic Optimality Laws: ② PSSH → the intermediate concentration is going to zero

Elementary Mode 1

Mass balance on m_1 and m_2 :

$$\frac{dm_1}{dt} = r_1 - r_{G_1} - r_2' + r_1' \approx 0$$

$$\frac{dm_2}{dt} = r_2' - r_1' - r_{G_1} \approx 0 \Rightarrow r_{G_1} = r_2' - r_1'$$

$$r_1 - (r_2' - r_1') - r_2' + r_1' = 0$$
$$\Rightarrow r_1 - r_2' + r_1' - r_2' + r_1' = 0$$
$$r_1 - 2r_2' + 2r_1' = 0$$
$$\frac{r_1}{2} = (r_2' - r_1')$$

$$\therefore \boxed{r_{G_1} = \frac{r_1}{2}}$$

Elementary Mode 2

$$\frac{dm_1}{dt} = r_1' - r_2' - r_{G_2} \approx 0 \Rightarrow r_{G_2} = r_1' - r_2' \Rightarrow -r_{G_2} = -r_1' + r_2'$$

$$\frac{dm_2}{dt} = r_2 + r_2' - r_1' - r_{G_2} \approx 0 \Rightarrow r_{G_2} = r_2 + r_2' - r_1'$$

$$\Rightarrow r_{G_2} = r_2 - r_{G_2}$$

$$\boxed{r_{G_2} = \frac{1}{2} r_2}$$

(iii) Mass balance E_1 and E_2 : $r_{e_1} = \frac{k_e' s_1}{K_e' + s_1}$; $r_{e_2} = \frac{k_e' s_2}{K_e' + s_2}$

$$\frac{de_1}{dt} = \alpha_1^* + \frac{k_E' S}{K_e' + s_1} u_1 - \beta_1 e_1 - r_{G_1} e_1$$

$$\frac{de_2}{dt} = \alpha_2^* + \frac{k_E' S}{K_e' + s_2} u_2 - \beta_2 e_2 - r_{G_2} e_2$$

α_1^* and α_2^* → 0 ; r_{G_1} and r_{G_2} → 0

$\therefore r_G = r_{G_1} + r_{G_2}$

Optimality laws:

$$u_1 = \frac{r_{G_1}}{r_{G_1} + r_{G_2}}$$

$$u_2 = \frac{r_{G_2}}{r_{G_1} + r_{G_2}}$$

$$\boxed{\frac{de_1}{dt} = r_{e_1} u_1 - \beta_1 e_1}$$

$$\boxed{\frac{de_2}{dt} = r_{G_2} u_2 - \beta_2 e_2}$$

$\beta_1 = \beta_2 = \beta_e$

$$\frac{dB}{dt} = r_{B_i} v_i$$

(iv) $\frac{ds_1}{dt} = -\frac{1}{Y_1} r_1 v_1$ $v_1 = \frac{r_{G_1}}{\text{Max}(r_{G_1}, r_{G_2})}$

$$\frac{dB}{dt} = r_{B_i} v_i$$

(iv) $\frac{ds_1}{dt} = -\frac{1}{Y_1} r_1 v_1$ $\qquad v_1 = \frac{r_{a_1}}{\text{Max}(r_{a_1}, r_{a_2})}$

$\frac{ds_2}{dt} = -\frac{1}{Y_2} r_2 v_2$ $\qquad v_2 = \frac{r_{a_2}}{\text{Max}(r_{a_1}, r_{a_2})}$

We also know, $r_1 = 2r_{a_1}$ $\qquad r_2 = 2r_{a_2}$

$$\frac{ds_1}{dt} = \frac{2 r_{a_1} v_1}{Y_1} = -\frac{1(2)}{Y_1} \frac{r_{a_1}^2}{\text{Max}(r_{a_1}, r_{a_2})}$$

$$\frac{ds_2}{dt} = \frac{2 r_{a_2} v_2}{Y_2} = -\frac{1(2)}{Y_2} \frac{r_{a_2}^2}{\text{max}(r_{a_1}, r_{a_2})}$$

Mass balance on B:

$$\frac{dB}{dt} = r_{a_1} v_1 + r_{a_2} v_2$$

$$r_1 = \frac{k_1 s_1 e_1}{K_1 + s_1} = 2r_{a_1}$$

$$r_2 = \frac{k_2 s_2 e_2}{K_2 + s_2} = 2r_{a_2}$$

(v) Cybernetic variables u_i and v_i expressed earlier

$u_1 = \frac{r_{a_1}}{r_{a_1} + r_{a_2}}$ $\qquad v_1 = \frac{r_{a_1}}{\text{Max}(r_{a_1}, r_{a_2})}$

$u_2 = \frac{r_{a_2}}{r_{a_1} + r_{a_2}}$ $\qquad v_2 = \frac{r_{a_2}}{\text{Max}(r_{a_1}, r_{a_2})}$

Respective relationships for r_{g_i} and r_{a_i} can be plugged in

(vi) Code attached.